

Coupling Billow to the trim

The minimum-energy structural model driven against the VSM quasi-steady trim

Integration, the modelling inputs it exposed, and first results

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Abstract

Billow — the standalone minimum-energy structural library documented in `billow.pdf` — is now driven against the VSM quasi-steady trim inside AWETrim’s existing coupled loop. This note covers the integration itself, three properties of the coupled problem that had to be understood before the answer was physical, two modelling inputs that were simply wrong, and the first converged results on the TU Delft V3.

The headline structural result: under trimmed aerodynamic load the wing loses 2.96 % of its tip-to-tip span, and it does so by *arc*ing rather than stretching — the leading-edge arc length is conserved to 0.21 % while the arch rise grows 2.4 %. The six main struts stay effectively straight (22–44 mm of sagitta over 2.2–2.6 m), which is the behaviour a real inflatable strut has and which the model only reproduces once the canopy is given its real stiffness.

A Markdown version of this document, kept in step with it, is `docs/billow/integration.md`.

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1 Architecture

The coupling adds one module and reuses everything else.

<code>aerostructural/fem/aerostructural_coupled_solver.py</code>	the coupled loop, SHARED
<code>aerostructural/fem/read_struct_geometry_yaml.py</code>	the geometry reader, SHARED
<code>aerostructural/fem/aero2struc.py</code>	the load mapping, SHARED
<code>aerostructural/aerodynamic_vsm.py</code>	the trim, SHARED
<code>aerostructural/billow/structural_billow.py</code>	new — the only new physics-facing code
<code>scripts/aerostructural/run_simulation_BILLOW.py</code>	new — the driver script

The coupled solver in `fem/` is, despite its location, not FEM-specific: it already dispatched on `config["structural_solver"]` between `pss` and `kite_fem`. Adding Billow was a branch at each of seven dispatch points, not a copy of its thousand lines. **Any future backend should do the same.**

`structural_billow.py` is the only module that imports both `awetrim.structural` and the AWETrim schema. That boundary is deliberate and load-bearing: the structural package stays free of PSS, VSM and YAML knowledge, so its element physics can go on being validated against closed-form solutions independently of anything the coupling does.

1.1 The coupled loop

Unchanged from the FEM path except for step 7:

1. structural nodes → VSM leading/trailing edge points
2. `body_aero.update_from_points`, bridle-line drag segments rebuilt
3. VSM quasi-steady trim → panel forces, moments, trim state
4. panel loads → structural nodes (`aero2struc`, §4)
5. inertial and gravity loads distributed by nodal mass
6. bridle line drag
7. `structural_billow.run_billow` → new node positions
8. Aitken relaxation on the node displacement
9. tape actuation, if any
10. convergence check

1.2 Element mapping

`structural_billow.instantiate` consumes exactly the arrays `read_struct_geometry_yaml.main` already returns, so one reader serves both backends.

reader element	Billow element	note
<code>inflatable_beam</code>	<code>InflatableBeamKernel</code>	diameter in <code>k_arr</code> , pressure in <code>c_arr</code>
<code>pulley arm pairs</code>	<code>PulleyKernel</code>	rest length is the WHOLE rope
<code>bridle noncompressive</code>	<code>CableKernel</code>	tension-only
<code>canopy grid springs</code>	dropped	replaced by wrinkling CST membrane triangles

Replacing the canopy is the point of the exercise. The FEM canopy is a square spring net with diagonals: too soft by $1 - \nu$ in tension, carrying shear only as a fourth-order effect of the diagonal stretch, and with no notion of a biaxial stress state at all. A relaxed (Pipkin) membrane carries the fabric's real biaxial law and settles into a tension field rather than mesh-scale crumple.

Everything else — bridle, pulleys, tubes — is element-for-element the same as the FEM model, so a Billow-versus-FEM difference is attributable to the canopy and the solver formulation and to nothing else.

On `struc_geometry_FEM_full.yaml`: **268 nodes, 1098 DOF, 47 cables, 20 pulleys, 97 inflatable tube beams, 378 membrane triangles**, replacing 693 canopy springs. Build under a second; the structural solve costs 2–5 s per coupled iteration against 8–25 s of VSM, so the structure is not the bottleneck.

1.3 Geometry

Billow requires `struc_geometry_FEM_full.yaml`. It is the only file in the repository carrying `strut_tubes`, `leading_edge_tubes` and `pressure`; the reduced PSM files are bridle-and-spring only and cannot feed beams or a canopy mesh.

That choice has a consequence worth stating plainly. FEM_full has 28 leading-edge nodes, hence 27 sections, so the reachable aero panel counts are $27n$: **27, 54, 81, 135**. The wes-quasi-steady campaigns run 45 panels, but that is 9×5 on the PSM geometry and is *not reachable here*. 54 is the adopted default — nearest to 45 and above the 27/36/45 band `as_config` records as mesh converged.

2 Three things that had to be true

Each was found by looking at the converged shape and asking whether it could be right. None is a solver bug; all three are properties of the coupled problem that were invisible in the FEM path.

2.1 The bridle is not in equilibrium as authored

The kite YAMLs store *measured* rest lengths against *measured* node positions, and the two do not agree. On FEM_full, `Br_main_1` is 11.5 % longer than the distance between the nodes it joins — at the real EA/l_0 that is 88 kN **in one line**. Starting a coupled run there is starting it from an explosion.

`relax_bridles` settles the bridle onto the held wing before the model is built: every wing grid node fixed, the KCU pulled down and then settled, the whole structure re-centred. Peak cable tension $88 \text{ kN} \rightarrow 0.4 \text{ N}$. Because the wing is held, the canopy and tube reference configurations are untouched up to a rigid translation, which no reference strain can see.

This is the same role `structural_kite_fem.relax_bridles` plays on the FEM path. **The consequence for callers: the nodes the coupled loop starts from are not the raw YAML nodes** — read them back from `structure.model.nodes`.

Related: the reader used to compute the bridle stiffness as EA/l_0 and then discard it for a hard-coded 5000 N m^{-1} , a bridle roughly $17\times$ too soft. Fixed; the PSS reader had always done it correctly.

2.2 Reference curvature breaks before the physics does

The beam's reference curvature is $\omega_0 = \psi/L_0$ with ψ a Rodrigues vector. It grows like $\tan(\theta/2)$ and is singular at $\theta = \pi$, so a *short* element bridging a *large* frame change is numerically hostile long before it is physically wrong.

Seeding each member's roll independently — the obvious thing — put the worst LEI-V3 element at 173° , $\omega_0 = 753 \text{ m}^{-1}$ **on a 17 mm element**, seven degrees from the chart singularity.

The fix uses a property of the section rather than a numerical trick. An inflated tube is isotropic in roll ($GA_2 = GA_3$, a single bending law), so only $\mathbf{d}_1$ is physically determined and the roll is entirely free. Spending it on conditioning — transporting frames by minimal rotation over the beam tree, and letting the *long* member own each leading-edge/strut joint

(`junction_frame="strut"`: the LE runs are 300–750 mm, the strut stubs 17–97 mm) — gives 10.6 m^{-1} , a factor of 71.

2.3 A gravity-only load case is not a smoke test

The KCU is pinned *below* the wing, so gravity alone slackens every bridle line and the bridle knots become a mechanism. The residual then sits at exactly their weight — 2.021 N on nodes 79 and 113 — and no amount of solver tuning moves it.

Billow reports this faithfully. `kite_fem` never shows it because its `I_stiffness = 25` acts as a ground spring on every node, quietly holding unsupported nodes up. **Read a stuck residual equal to some node’s load as “this node has no load path”, not as a conditioning problem.**

3 Two inputs that were wrong

3.1 Apparent wind speed, and the end of self-similarity

`as_config`’s flight state is zero elevation, zero azimuth, 90° course, gravity off — the *fastest corner of the wind window*. At 8 m s^{-1} wind and the trimmed L/D of about 4.8 that is $v_a \approx 38 \text{ m s}^{-1}$, above the entire $13\text{--}25 \text{ m s}^{-1}$ band the wes-quasi-steady campaigns sweep.

For the FEM path this mattered less than it looks, because the campaigns argue that coupled trims are v^2 -self-similar and correct for it. **That argument does not survive the introduction of the tubes.** The inflatable bending law

$$M(\kappa) = M_{\max} \left(1 - e^{-EI_0 \kappa / M_{\max}} \right) \quad (1)$$

saturates. Past $M_{\max}$ the tube has essentially no incremental bending stiffness and hinges at constant moment — the compression-side wrinkling collapse of a real inflated tube. Collapse is reported, not enforced, because a dropping post-collapse moment would be an unbounded mechanism that energy minimisation would simply run away from.

A saturating law has no self-similarity: the load decides whether the tubes are inside their calibrated range at all. At $v_a = 38 \text{ m s}^{-1}$ every strut sat 3.0–4.7 **times past its collapse curvature**, at 86–94 % of $M_{\max}$, and hinged into 17–21 % of chord.

3.2 Canopy stiffness

The canopy was initially set to $Et = 5000 \text{ N m}^{-1}$ to match the `kite_fem` spring net, on the reasoning that it made the two canopies comparable. It made them comparable *and both wrong*: at any plausible sailcloth thickness 5000 N m^{-1} is $E \approx 20 \text{ MPa}$, which is rubber.

The real cloth is already described in the geometry file. `canopy_density`: 170 g m^{-2} of polyester ripstop gives a solid-equivalent thickness of $0.170/1380 = 123 \mu\text{m}$, and at a crimp-derated in-plane modulus of 4 GPa that is $Et \approx 4.9 \times 10^5 \text{ N m}^{-1}$ — about a hundred times stiffer.

A canopy that soft does not carry load; it billows, and drags the struts with it. Measured on the same load case, strut sagitta ran 4.5–6.7 % of chord at $5 \times 10^3 \text{ N m}^{-1}$ and 0.6–4.1 % at $1 \times 10^6 \text{ N m}^{-1}$.

The modulus is a **material estimate, not a measurement**. The repository contains no tensile data for this cloth. The plausible range — 2–12 GPa fibre modulus, varying crimp derating — spans Et from 2.5×10^5 to $1.5 \times 10^6 \text{ N m}^{-1}$. A tensile test on the actual canopy would replace it, and should.

4 Chordwise load placement is the local pitching moment

Independent of Billow, but surfaced by it, and it changes the FEM path too.

`aero2struc` spreads each spanwise VSM panel force over ten chordwise nodes. **Where it puts the resultant is the panel's local pitching moment**, and the wing's trim is a moment balance — so this is a first-order modelling choice, not a detail of interpolation.

The historical `cp_file` mode reads one measured ΔC_p shape from a single-angle file and applies it to every panel at every angle of attack. Its centroid on the LEI-V3 file is $0.291c$. Measured on one trim at 135 panels, the panels' actual centres of pressure — which VSM already computes from each panel's own force and moment — run $0.303\text{--}0.805c$. The fixed station therefore imposes 917 N m of spurious pitching moment, about -10 N m on nearly every panel *with the same sign*. A systematic bias is precisely what shifts a moment balance; a random one would largely cancel.

`chordwise_distribution: moment_matched` keeps the measured shape as a prior and tilts it onto each panel's own centre of pressure,

$$w_i = \frac{w_i^0 e^{\lambda t_i}}{\sum_j w_j^0 e^{\lambda t_j}}, \quad \sum_i w_i t_i = t_{\text{cp}}, \quad (2)$$

solving the scalar equation for λ . Three properties make the exponential tilt the right correction rather than, say, a least-squares one: every weight stays strictly positive, so no unphysical negative patch of load appears; the weights still sum to one, so the panel force is preserved exactly; and λ is the natural parameter of an exponential family, whose mean is strictly increasing in it, so the root exists, is unique, and Newton converges from zero. It is also the minimum-relative-entropy correction, so the measured shape is kept wherever the moment does not contradict it.

Residual moment error: $917\text{ N m} \rightarrow 6 \times 10^{-9}\text{ N m}$. The default stays `cp_file` so stored results reproduce; the Billow script sets `moment_matched`.

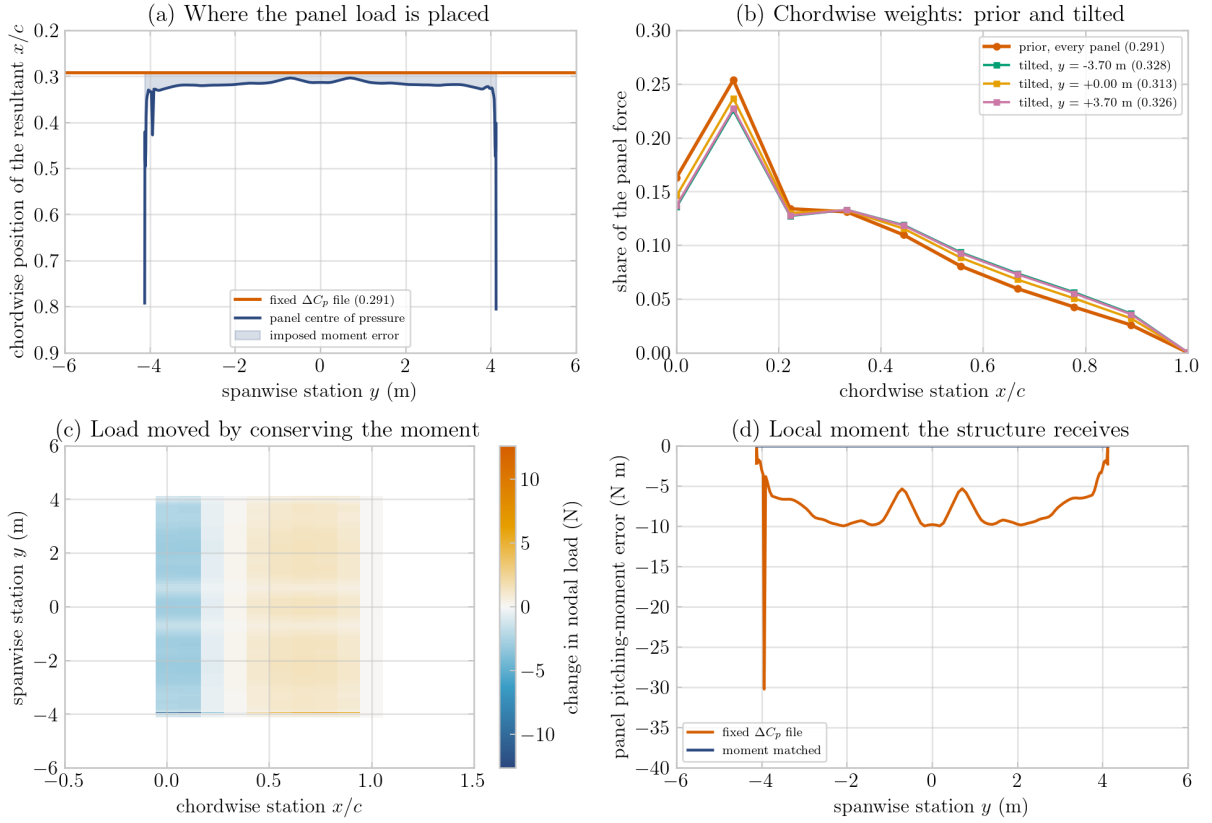


Figure 1: Conserving the panel pitching moment. (a) the single fixed chordwise station the ΔC_p file imposes against each panel’s actual centre of pressure — the shaded gap is moment the structure receives but the aerodynamics never produced; the excursions at $y \approx \pm 4$ m are the stalled tip panels. (b) the prior and three tilted weight sets. (c) the load the correction moves: forward of $0.2c$ and aft into 0.4 – $0.9c$. (d) the resulting per-panel moment error, about -10 N m on nearly every panel with the same sign, which is what makes it bias a trim, against zero for the matched weights.

A warning about the diagnostic. `check_moment_preservation` does *not* measure any of the above. Both of its call sites hand it the already-distributed loads, so it reports the error of the *spatial* nearest-node step alone. That error is 0.09% on the built shape but around 20% once the kite has deformed — a separate and still-open defect. Do not quote its numbers as evidence about the chordwise placement.

5 Preliminary results

LEI-V3, unactuated (the tape lengths the geometry stores: `depower_tape` $l_0 = 1.9$ m, i.e. $u_{dp} \approx 0.28$), 54 aero panels, gravity off, moment-matched chordwise loads, $v_a \approx 24$ m s $^{-1}$.

5.1 Convergence

state	iterations	final residual
real canopy, v_a 24	2	3.84 N
soft canopy, v_a 38	7	4.33 N
soft canopy, v_a 38, 135 panels	10	4.45 N

Gate is 5 N absolute. The physically correct case is also by far the easiest to converge — worth noting, because a model driven outside its calibrated range does not merely give a wrong answer, it costs more to get it.

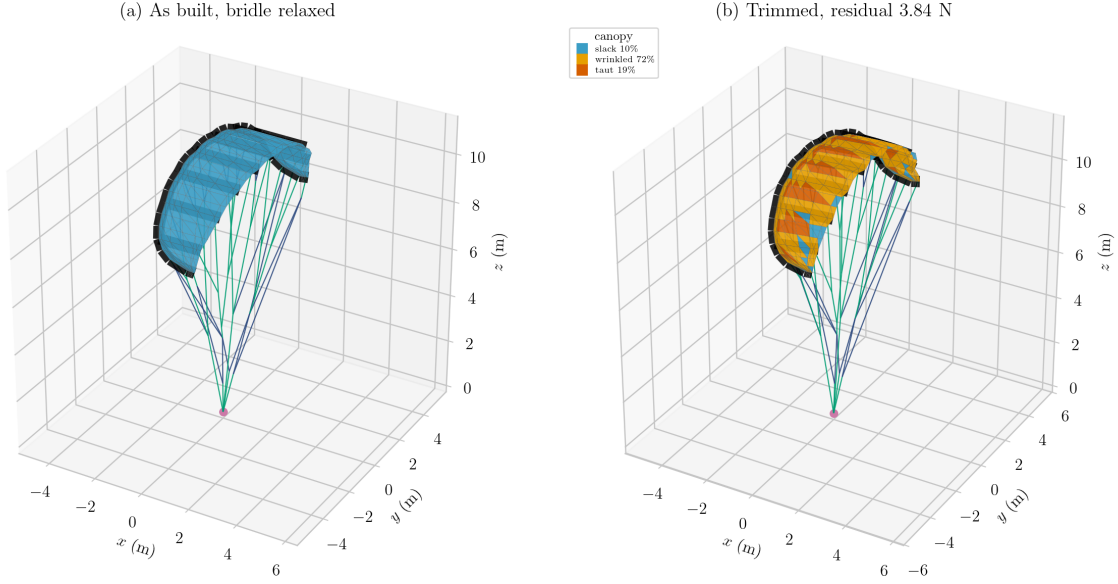


Figure 2: The first converged coupled solve. **(a)** the built shape after the bridle relaxation of §2.1, before any aerodynamic load; **(b)** the trimmed shape, canopy shaded by the wrinkling state the solver chose. Black: inflatable tubes, drawn at their real relative diameters. Green: bridle cables. Blue: pulley ropes through their sheaves. The magenta marker is the KCU, the only pinned node.

5.2 Span reduction

Tip-to-tip is the straight distance between the two leading-edge tips; LE arc is the length measured along the tube.

	as built	trimmed	change
tip-to-tip span	8.202 m	7.959 m	−0.243 m, −2.96 %
leading-edge arc length	12.662 m	12.636 m	−0.026 m, −0.21 %
arch rise above the tip chord	3.282 m	3.362 m	2.42 %
mean chord	2.044 m	2.036 m	−0.4 %

The span reduces by 3.0 %, and it does so by arcing, not by stretching. The leading-edge arc length is conserved to 0.21 % — the tube neither extends nor compresses — while the arch rise grows 2.4 %. The wing bends into a deeper arc under load and pulls its tips inward. That is the correct physical picture for a leading-edge-inflatable kite, and it is a stringent check on the beam formulation: an axially sloppy beam would have shown span loss through shortening instead.

For contrast, the same kite with the soft canopy at $v_a = 38$ gives the **wrong sign**: tip-to-tip 2.24 %, arch rise −3.87 %, mean chord −7.1 %, LE arc unchanged. A canopy at 20 MPa stretches instead of carrying load, so the sail draws the trailing edge toward the leading edge while the wing *flattens* and splays its tips outward. Both the sign of the span change and the magnitude of the chord change are unphysical. This is the clearest single argument for §3.2.

5.3 Strut bending

Sagitta is the maximum perpendicular deviation from the straight line joining the strut’s two ends. Utilisation is curvature over the law’s collapse curvature; above 1.0 the tube is outside the range the ASKITE fits were calibrated over.

configuration	main struts (6)	outboard struts (2)
v_a 38, canopy 5 kN m^{-1}	17–21 %, $3.0\text{--}4.7\times$	—
v_a 21, canopy 5 kN m^{-1}	7.1–9.5 %, $1.2\text{--}2.2\times$	—
v_a 24, canopy 493 kN m^{-1}	1.0–1.8 % ($22\text{--}44 \text{ mm}$), $0.17\text{--}0.25\times$	6.6–6.8 %, $1.6\times$

The six main struts are effectively straight — tens of millimetres over 2.2–2.6 m — and sit comfortably inside the calibrated range. This matters because real LEI struts do not visibly bend in flight, and a model that bends them is wrong whatever its residual says.

The two outboard struts remain past collapse. They are the thinnest (0.080–0.106 m against 0.108–0.141 m) and the shortest, and they sit next to the tip region where VSM reports stall, so they are where the inflation pressure matters most.

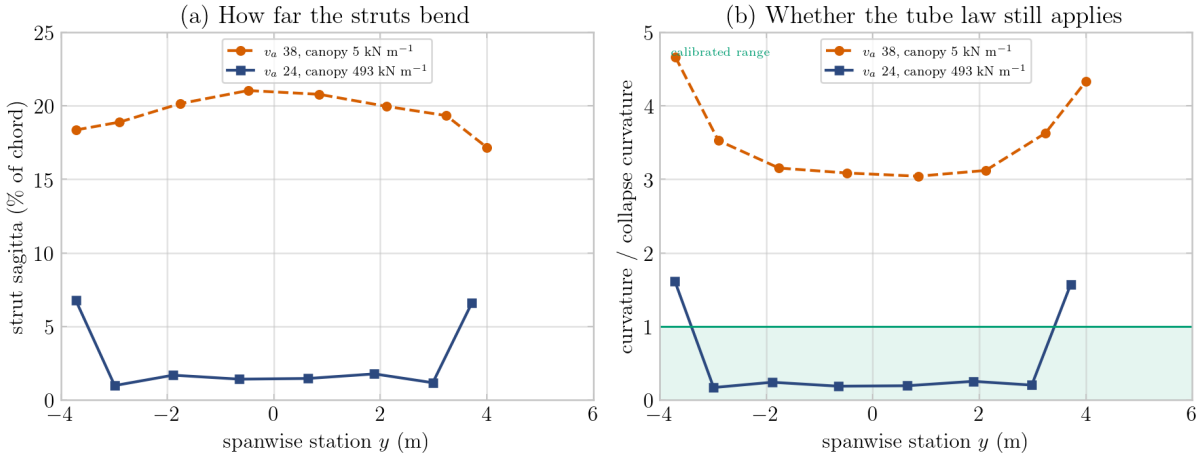


Figure 3: Strut bending along the span. **(a)** sagitta as a fraction of chord; **(b)** curvature over the collapse curvature of the fitted tube law, with the calibrated range shaded. At v_a 38 with the spring-net canopy every strut is far outside the range the ASKITE fits cover, so the saturated moment leaves them hinging. With the real apparent speed and the real cloth the six main struts fall to a fifth of collapse curvature and bend by tens of millimetres; only the two thin outboard struts, next to the stalled tip, remain above 1.

5.4 Canopy state

Fraction of the 378 membrane triangles in each regime at the solved state.

	slack	wrinkled	taut	resultant aero load
real canopy, v_a 24	9.5 %	72.0 %	18.5 %	3.57 kN
soft canopy, v_a 38	1.9 %	41.0 %	57.1 %	10.73 kN

The realistic canopy is predominantly **wrinkled** — carrying tension in one direction and none across it. That is the expected state for a sail: a genuine tension field over most of the surface, going taut only where it is pulled biaxially. The soft canopy at three times the load reads as mostly taut simply because it has stretched far enough to carry biaxial tension everywhere, which is what a rubber sheet would do.

This diagnostic has no counterpart in the spring-net canopy. A net has no biaxial stress state and so cannot distinguish these cases at all.

5.5 Depower sweep

Continuation chains either side of the built tape length, at three winds: 26 converged points over $u_{dp} = 0.243 - -0.314$, in 19 minutes. Each chain walks the tape inside a single coupled call so

every step warm-starts the next; the alternative, solving each point independently, was measured at 14 minutes *per point*.

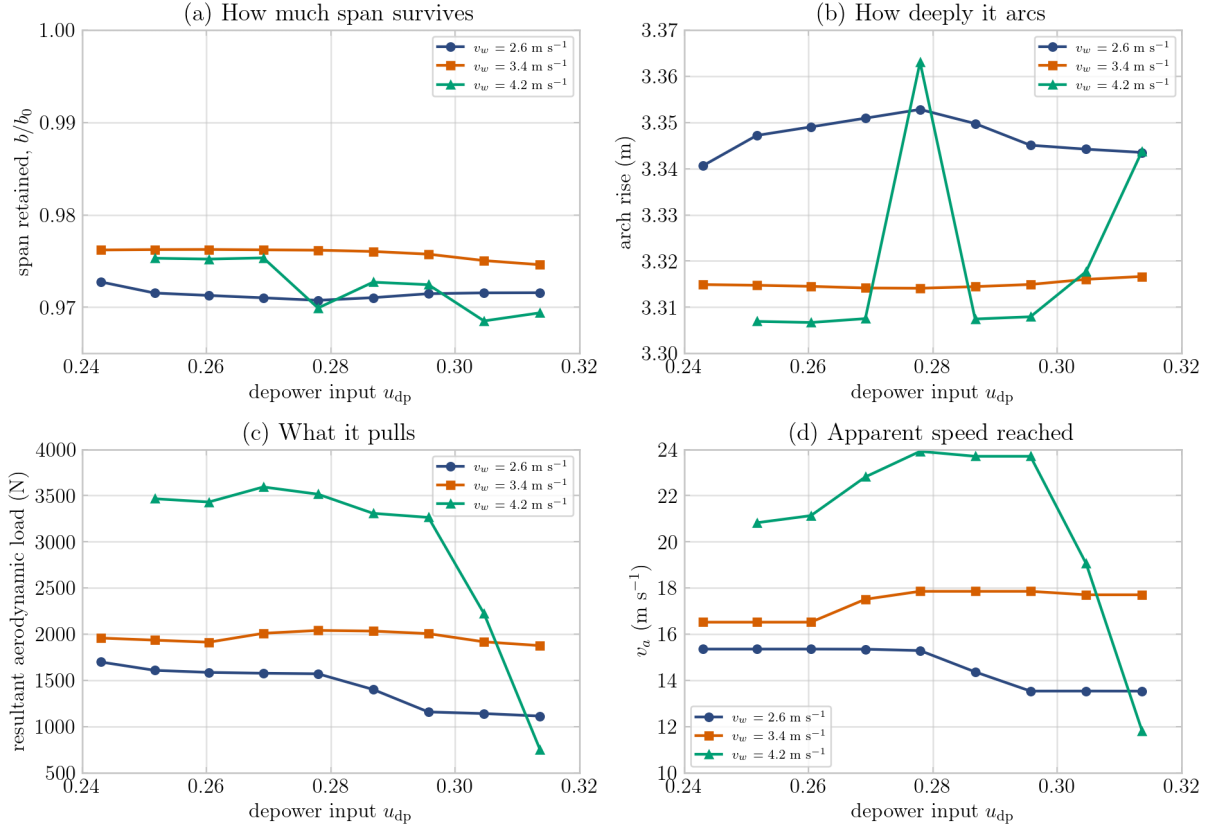


Figure 4: Depower chains at the window centre. Each curve is one wind speed; within a chain the wind is held, so v_a slides with u_{dp} and panel (d) is a readout rather than a control. (a) span retained against the as-built 8.202 m. (b) arch rise — the outlier at $u_{dp} = 0.278$ is the built point, the only step in each chain without a warm start. (c) resultant aerodynamic load. The $v_w = 4.2 \text{ m s}^{-1}$ curve leaves the attached branch past $u_{dp} \approx 0.30$ in both (c) and (d).

Span loss is nearly invariant over the box. Between 96.9 % and 97.6 % of the built span is retained everywhere in the grid: depowering by 0.07 in u_{dp} and nearly doubling the wind barely move it. The 2.96 % of $\$5.2$ is therefore the wing’s signature across this operating box rather than a property of one trim point.

What depower controls is load, not shape. The resultant aerodynamic force falls 29 % at $v_w = 2.6 \text{ m s}^{-1}$ and 35 % at 3.4 m s^{-1} across the u_{dp} range, against about 0.5 % of span movement. At least this far from stall, the depower tape is a load knob.

Three caveats, all visible in the figure and none yet resolved:

1. **The highest-wind chain leaves the attached branch past $u_{dp} \approx 0.30$:** load 3263–751 N, v_a 23.7–11.8 m s^{-1} . At $u_{dp} = 0.314$ the highest wind pulls *less* than the lowest, which is deep stall rather than depower. Those points want the attached-branch guard the campaign scripts use.
2. **The built point of each chain is under-converged,** being the one step with no warm start (4.11 N against about 2 N); it is the arch-rise outlier in panel (b). An artifact of chain structure, not physics.
3. **Span is not monotonic in wind** — 2.6 m s^{-1} loses more span than 3.4 m s^{-1} . Unexplained; it may be caveat 2 leaking in.

The flat v_a segments in panel (d) are `quasi_steady_trim.max_nfev: 8`: the warm-started trim

returns a bit-identical `opt_x` when it starts inside tolerance, so v_a repeats exactly. Harmless, but v_a per step is not independently converged.

6 Open items

1. **Strut inflation pressure.** `struc.geometry_FEM_full.yaml` records `pressure: 0.3 bar`, which sets both $M_{\max}$ and EI_0 through the ASKITE fits. The two outboard struts are still $1.6\times$ past collapse, and being the thinnest they are the most pressure-sensitive members on the kite. The real V3 operating pressure should be confirmed.
2. **Canopy modulus is an estimate** (§3.2). Its plausible range is roughly $\pm 3\times$, and the results are sensitive to it — including the *sign* of the span change at the soft end.
3. **The spatial load mapping loses moment.** `map_aero_forces_to_struct_nodes` drops $\sim 20\%$ of the applied moment once the kite has deformed (§4). The chordwise placement is now exact; this step is not.
4. **Tube axial and shear stiffness are derived, not fitted.** The ASKITE fits cover bending and torsion only; EA and κGA are read off the same fits' initial slopes as a modulus over the thin-wall section. Defensible and not a sensitive quantity, but an assumption.
5. **The stiffness ladder is untested end to end.** `set_stiffnesses` exists and is unit-tested, but no coupled run has used a stiffness continuation ramp the way the PSS path does.
6. **Steering and depower actuation are wired but unexercised.**

7 Running it

```
python scripts/aerostructural/run_simulation_BILLOW.py
```

Results land in `results/<kite>/aerostructural/billow/<case>/` as `sim_output.h5` plus the deformed structural and aerodynamic geometry YAMLS. The results directory is keyed on actuation only and is **overwritten** between runs that differ in anything else — copy a run aside before changing panel count, wind or canopy stiffness.

Numerics live in the `structural_billow` block of `as_config.yaml`; every key is listed with its default in `structural_billow.DEFAULTS`, and unknown keys are rejected rather than ignored. Two diagnostics:

```
python scripts/aerostructural/plot_billow_geometry.py
python scripts/aerostructural/plot_chordwise_moment_matching.py
```

A depower sweep is a pair of continuation chains per wind, walking the tape either side of its built length and converging at every step:

```
python scripts/aerostructural/run_chain_depower_BILLOW.py --wind 2.6 3.4 4.2
python scripts/aerostructural/plot_billow_depower_chains.py
```

That writes `chains.csv` and one `sim_output.h5` per chain under `results/<kite>/aerostructural/billow_depower_`. Each row carries a `converged` flag: a step that missed the residual gate is recorded rather than dropped, so a short chain cannot read as a complete one. `run.sweep_depower_BILLOW.py` solves each point independently instead and iterates the wind onto a target apparent speed — about twenty times more expensive, and worth it only when a few points must sit at an exact v_a .

`sim_output.h5` now stores the nodal material frames alongside the positions (`setup_tracking_arrays(..., with_frames=True)`). A geometrically exact beam's curvature lives in the frames, not in the node positions, so without them a saved run cannot afterwards be asked how bent its tubes were or whether they had passed collapse — which is exactly the question the strut results turn on.